

1. Intuitive Concept of Perceptron Learning Algorithm:

- A. adjustment of bias weight (w_0) depends on desired output value " y " in pair training examples.
therefore, if the majority of output value " y " in pair training examples are positive / negative,
then adjustment of bias weight (w_0) will approach (tend to) positive / negative.
- B. only weight (w_i) that can significantly affect predicted output value " P_y " (= corresponding input (x_i) is larger) will see significant changing on adjustment of weight (w_i).
in extremely, the weights (w_i) will see 0 adjustment when corresponding input (x_i) is 0.

2. Geometric Understanding of Two-Input Perceptron:

for $z = w_0 \cdot x_0 + w_1 \cdot x_1 + w_2 \cdot x_2$ the equation can be seen as a plane for a two-input perceptron in 3D space,
because $w_1 \cdot x_1 + w_2 \cdot x_2 + (-1) \cdot z + w_0 \cdot x_0 = 0 \implies z = w_0 \cdot x_0 + w_1 \cdot x_1 + w_2 \cdot x_2$,
($a \cdot x + b \cdot y + c \cdot z + d = 0$ plane in 3D space)